19 Burchaklar va Uchburchaklar

1-to'plam. *To'ldiruvchi, yondosh va vertikal burchaklar.*

1 Two angles are complementary. One measures 37°. Find the other.

2 Two angles are supplementary. One measures 112°. Find the other.

3 Two angles are vertical angles. One measures 74°. Find the other.

4 Two complementary angles measure $2x$ and $3x$. Find x, then find both angles.

5 Two supplementary angles measure $4x + 10$ and $2x + 20$. Find x, then find both angles.

6 Two vertical angles measure $3x + 15$ and $5x - 25$. Find x, then find the angle.

2-to'plam. *Uchburchak ichki burchaklari yig'indisi.*

1 A triangle has angles 50° and 65°. Find the third angle.

2 A triangle has angles 90° and 35°. Find the third angle.

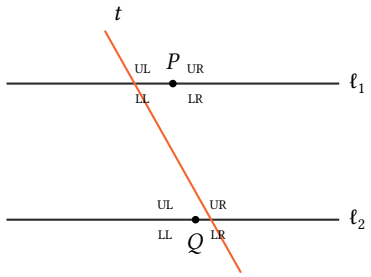
3 An equilateral triangle has three equal angles, each x. Find x.

4 A triangle's angles measure $2x$, $3x$, and $4x$. Find x, then find each angle.

5 A triangle's angles measure $x + 10$, $x + 20$, and $x + 30$. Find x, then find each angle.

6 An isosceles triangle has two equal base angles, each $2x$, and a vertex angle of $x + 60$. Find x, then find all three angles.

3-to'plam. *Parallel to'g'ri chiziqlar va kesuvchi.* $\ell_1 \parallel \ell_2$, *kesuvchi* t *ularni* P *va* Q *nuqtalarda kesadi.*



1 P's lower-right angle is 72°. Find Q's upper-left angle.

2 P's lower-left angle is 96°. Find Q's upper-right angle.

3 P's lower-right angle is 130°. Find Q's upper-right angle.

4 P's upper-right angle is 108°. Find Q's upper-right angle.

5 P's lower-left angle is 85°. Find Q's lower-left angle.

6 Lines m and n cut by a transversal, $m \parallel n$ is *not* stated. One angle is 85°, the alternate-interior angle is 95°. Can you conclude $m \parallel n$?

4-to'plam. *O'xshash uchburchaklar: harflar tartibiga rioya qiling.*

1 $\triangle ABC \sim \triangle DEF$. $AB = 4$, $BC = 6$, $AC = 8$, $DE = 6$. Find EF and DF.

2 $\triangle PQR \sim \triangle LMN$. $PQ = 5$, $QR = 12$, $PR = 13$, $LM = 15$. Find MN and LN.

3 $\triangle XYZ \sim \triangle ABC$. $XY = 7$, $YZ = 9$, $XZ = 11$, $BC = 18$. Find AB and AC.

4 $\triangle GHI \sim \triangle KLM$. $GH = 10$, $HI = 14$, $GI = 16$, $KL = 25$. Find LM and KM.

5 $\triangle RST \sim \triangle UVW$, scale $\frac{3}{2}$ (RST→UVW). $RS = 8$, $ST = 10$, $RT = 14$. Find UV, VW, UW.

6 $\triangle ABC \sim \triangle DEF$. $AB = 6$, $BC = 8$, $AC = 10$ (perimeter 24), $DE = 9$. Find the perimeter of $\triangle DEF$.

5-to'plam. *Aralash: burchaklar + o'xshashlik.*

1 In $\triangle ABC$, $\angle A = 50°$, $\angle B = 70°$. Find $\angle C$. Also $\triangle ABC \sim \triangle DEF$, $DE = 2 \times AB$; if $AB = 6$, find DE and state $\angle D$ (explain why, no new computation).

2 Two triangles have $\angle A = \angle D = 65°$, $\angle B = \angle E = 80°$. Are they similar? By which criterion? Find $\angle C, \angle F$.

3 $\triangle ABC \sim \triangle DEF$ by SAS: $AB = 8$, $DE = 12$, $\angle A = \angle D = 40°$, $AC = 10$. Find DF.

4 In $\triangle ABC$, D on AB, E on AC, $DE \parallel BC$. $AD = 4$, $DB = 6$, $AE = 6$. Find EC.

5 A 1.8 m person casts a 2.4 m shadow. A flagpole casts a 9.6 m shadow. Find the flagpole's height.

6 Two triangles both have angles $55°, 60°, 65°$. First triangle's side opposite $55°$ is 5; second's is 10. Are they similar? Find the scale factor.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $\ell_1 \parallel \ell_2$. P's lower-right angle is $70°$. Find Q's upper-left angle.

"Bular bir tomonlama ichki burchaklar: $180 - 70 = 110°$*."*

Xato va to'g'ri javob: .....

2 Uchburchak burchaklari $2x, 3x, 4x$. x ni toping.

"$2x + 3x + 4x = 360$, $9x = 360$, $x = 40$*."*

Xato va to'g'ri javob: .....

3 $\triangle ABC \sim \triangle DEF$. $AB = 6, BC = 9, AC = 12, DE = 8$. EF ni toping.

"$EF = AC \times \frac{DE}{AB} = 12 \times \frac{8}{6} = 16$*."*

Xato va to'g'ri javob: .....

4 Ikki burchak komplementar. Biri $65°$. Ikkinchisini toping.

"Yig'indi $180°$*:* $180 - 65 = 115°$*."*

Xato va to'g'ri javob: .....

5 $\ell_1 \parallel \ell_2$. P's upper-right (mos burchak) $112°$. Q's upper-right?

"Mos burchaklar yig'indisi $180°$*:* $180 - 112 = 68°$*."*

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – ko'p qadamli masalalar.*

1 In $\triangle ABC$, $\angle A = 3x$, $\angle B = 2x+10$, $\angle C = x+20$. Find x and all three angles, then classify as acute, right, or obtuse.

2 $\ell_1 \parallel \ell_2$. P's lower-left angle is $4y+10$. Q's upper-left is $6y-50$ (co-interior pair). Find y, then both angles.

3 $\triangle ABC \sim \triangle DEF$. Perimeter of ABC is 30, $AB : BC : AC = 2 : 3 : 5$. $DE = 10$. Find the perimeter of DEF (and EF, DF).

4 An isosceles triangle has vertex angle $3x - 10$ and each base angle $x + 15$. Find x, all three angles, and the exterior angle at a base angle.

5 A 3 m stick casts a 4 m shadow. A tree casts a 16 m shadow. (a) Find the tree's height. (b) Find the straight-line distance from the top of the tree to the tip of its shadow.

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	53°	2	68°
	3	74°	4	$x = 18; 36^\circ, 54^\circ$
	5	$x = 25; 110^\circ, 70^\circ$	6	$x = 20; 75^\circ$
2-to'plam	1	65°	2	55°
	3	$x = 60^\circ$	4	$x = 20; 40^\circ, 60^\circ, 80^\circ$
	5	$x = 40; 50^\circ, 60^\circ, 70^\circ$	6	$x = 24; 48^\circ, 48^\circ, 84^\circ$
3-to'plam	1	72° (almashinuvchi)	2	96° (almashinuvchi)
	3	50° (bir tomonlama)	4	108° (mos burchak)
	5	85° (mos burchak)	6	Yo'q, $m \nparallel n$
4-to'plam	1	$\frac{3}{2}$; $EF = 9, DF = 12$	2	3; $MN = 36, LN = 39$
	3	2; $AB = 14, AC = 22$	4	$\frac{5}{2}$; $LM = 35, KM = 40$
	5	$UV = 12, VW = 15, UW = 21$	6	$\frac{3}{2}$; perimetr 36
5-to'plam	1	60°; $DE = 12, \angle D = 50^\circ$	2	Ha, AA; $35^\circ, 35^\circ$
	3	$DF = 15$	4	$EC = 9$
	5	7.2 m	6	Ha, AA; koeffitsiyent 2
6-to'plam	1. Xato – bular almashinuvchi ichki (teng), bir tomonlama emas. To'g'ri: 70°.			
	2. Xato – uchburchak burchaklari yig'indisi 180°, 360° emas. To'g'ri: $x = 20$.			
	3. Xato – mos tomon BC, AC emas. To'g'ri: $EF = 9 \times \frac{8}{6} = 12$.			
	4. Xato – komplementar yig'indisi 90°. To'g'ri: 25°.			
	5. Xato – mos burchaklar teng, yig'indi emas. To'g'ri: 112°.			
7-to'plam	1. $x = 25; 75^\circ, 60^\circ, 45^\circ$ – o'tkir uchburchak.			
	2. $y = 22$; $P = 98^\circ, Q = 82^\circ$.			
	3. AB=6, BC=9, AC=15; koeff. $\frac{5}{3}$; EF=15, DF=25; perimetr 50.			
	4. $x = 32$; uch 86°, asos $47^\circ, 47^\circ$; tashqi burchak 133°.			
	5. (a) balandlik 12 m (b) masofa = $\sqrt{12^2 + 16^2} = 20$ m.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

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